

Stochastic Execution and Cost-Risk Analysis

Adrián Vázquez

May 26, 2026

Abstract

We develop a stochastic optimal execution framework under market impact, comparing TWAP, VWAP, and Almgren–Chriss strategies. Using Monte Carlo simulation, we evaluate execution performance through implementation shortfall, expected cost, and execution risk, and analyze the cost-risk trade-off induced by execution urgency.

1 Introduction

Executing large orders introduces a trade-off between:

- Market impact (cost)
- Price uncertainty (risk)

Deterministic strategies fail to capture this interaction. We extend the analysis to a stochastic setting.

2 Stochastic Price Dynamics

We model prices as:

$$dS_t = \sigma dW_t \tag{1}$$

Discrete version:

$$S_{k+1} = S_k + \sigma\sqrt{\tau}\epsilon_k, \quad \epsilon_k \sim \mathcal{N}(0, 1) \tag{2}$$

Simulated Brownian Price Path

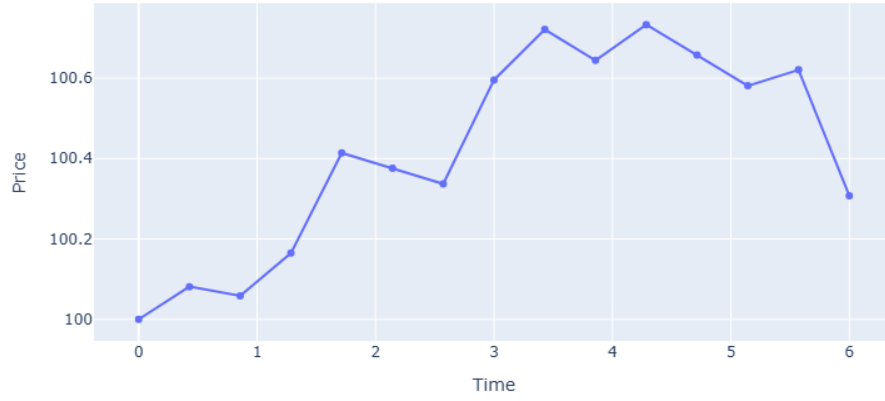


Figure 1: Single simulated price path

Insight: Price paths exhibit randomness, introducing uncertainty in execution outcomes.

Multiple Simulated Price Paths

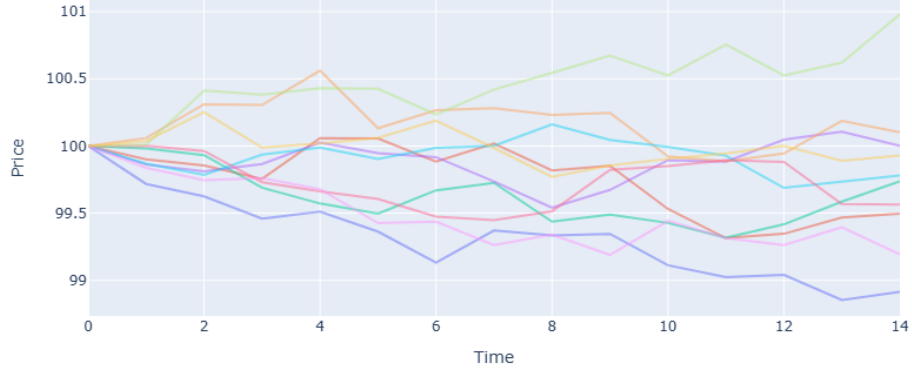


Figure 2: Multiple simulated price paths

Insight: Different realizations lead to different execution costs even under identical strategies.

3 Execution with Market Impact

Execution price:

$$P_k = S_k + \gamma \sum_{j < k} n_j + \eta \frac{n_k}{\tau} \quad (3)$$

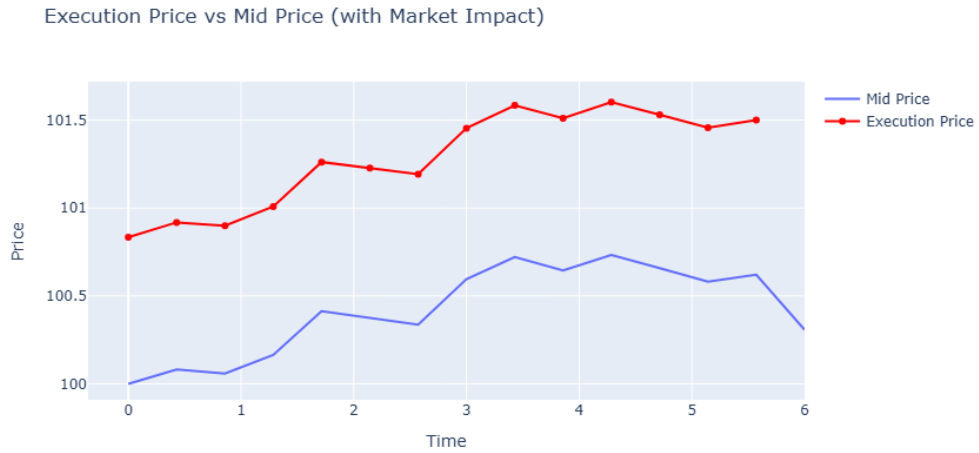


Figure 3: Execution vs Mid Price

Insight:

- Execution prices are systematically higher than mid prices
- Impact accumulates over time

4 Performance Metrics

4.1 Implementation Shortfall

$$IS = \sum_{k=1}^N n_k P_k - QS_0 \quad (4)$$

4.2 Example Output

Total Cost	Avg Price	IS per Share
64205.75	101.28	1.284

Insight: Execution cost emerges from both market impact and stochastic price movement.

5 Monte Carlo Evaluation

We simulate multiple price paths and evaluate strategies.

5.1 Strategy Comparison

Strategy	Mean Cost	Std Cost	Min Cost	Max Cost
TWAP	43114	16543	-2917	95952
VWAP	51045	16399	2552	101731
Almgren-Chriss	43151	16268	-1760	95105

Insights:

- VWAP shows higher expected cost and heavier tail risk
- TWAP and Almgren-Chriss have similar mean cost
- Almgren-Chriss slightly reduces execution variance

5.2 Distribution of Implementation Shortfall

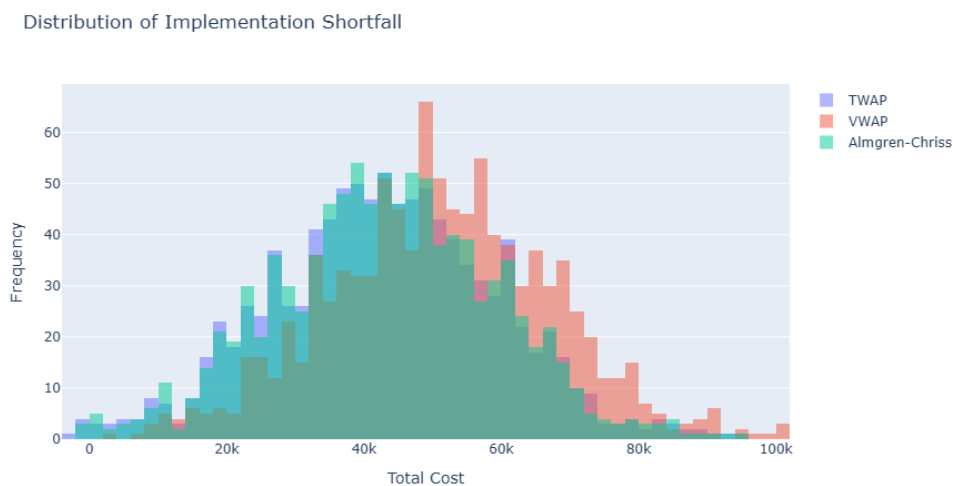


Figure 4: Distribution of Implementation Shortfall

Insight:

- Execution outcomes follow a distribution, not a single value
- VWAP has a heavier right tail (higher extreme cost risk)

6 Cost-Risk Frontier

By varying λ :

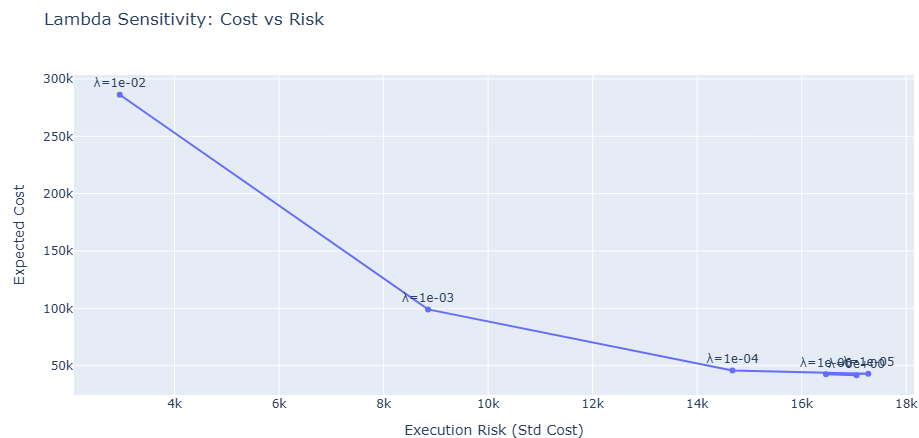


Figure 5: Cost-Risk Frontier

Insights:

- Low $\lambda \rightarrow$ low cost, high risk
- High $\lambda \rightarrow$ high cost, low risk
- Trade-off is non-linear

Key Insight:

Small increases in λ significantly reduce risk with limited cost increase.

7 Key Questions

1. **Price uncertainty:** introduces variability in execution outcomes.
2. **Market impact:** increases execution prices above mid.
3. **Best strategy:**
 - No universal winner
 - Almgren–Chriss dominates in flexibility
 - Optimal choice depends on λ

8 Conclusion

Execution is a stochastic control problem under market impact.

- Cost and risk must be evaluated jointly
- Distributional analysis is essential
- Almgren–Chriss provides a flexible framework

9 References

Almgren, R., & Chriss, N. (2001). Optimal execution of portfolio transactions.

Gatheral, J.& (2010). ****No-Dynamic-Arbitrage and Market Impact.**** *Quantitative Finance**, 10(7), 749-759.

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